

David Le

davidducle2@gmail.com | (617) 397-0049 | Boston, MA | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

University of Massachusetts Boston

B.S. in Computer Science | GPA: 3.78 / 4.00

Boston, Massachusetts

Expected May 2027

Relevant Coursework: Computer Security, Database Management, Intro to Software Engineering, Applied Machine Learning, Industry Applied AI Lab

TECHNICAL SKILLS

Languages & Query: Python, SQL, Java, JavaScript, C, HTML/CSS

Security: OAuth 2.0, Firebase Auth, role-based access control, Firestore security rules, Fernet encryption, Metasploit, Aircrack-ng

Data & Analytics: Pandas, SQL, feature engineering, walk-forward validation, paired bootstrap significance testing

Frameworks & Libraries: FastAPI, Flask, LangGraph, React, React Native, Node.js, scikit-learn, XGBoost, PyTorch

Databases: PostgreSQL, SQLAlchemy, Oracle, Firebase, ChromaDB, FAISS

Cloud, DevOps & Tools: Docker, Git/GitHub, pytest, Vercel, Render, Hugging Face Spaces, Gemini

PROJECTS

Receipt Automation (AutoTax) [\[Demo\]](#) [\[GitHub\]](#) | *Python, LangGraph, FastAPI, Gmail API, Firestore*

July 2025 – Sept 2026

- Turns receipt emails into an accountant-ready ledger; rebuilt the pipeline as an eight-node LangGraph state machine that triages, extracts, reconciles and persists each receipt, and pauses for human review when the rules cannot prove a field.
- Every stored record carries a provenance trace naming the source of each field, whether pattern match, sender domain, model or a person, and a validation step sends back any receipt whose arithmetic does not reconcile.
- Cut model calls from one per email to a single batched call, **8× faster** at identical benchmark accuracy, and encrypted stored OAuth refresh tokens with Fernet.

Multilingual Mindfulness Assistant [\[Demo\]](#) [\[GitHub\]](#) | *React Native, Firebase, Python, Gemini*

Feb – June 2026

- Shipped a multilingual mindfulness app for older Korean adults and caregivers under the guidance of a UMass Boston professor, pairing a React Native client with a Python backend for streaming chat and text-to-speech.
- Wrote the Firestore security rules and backend authorization: per-user ownership checks, field allowlists on every write, completed session records that cannot be edited or deleted, and server-issued tokens bound to the signed-in user so one account cannot reach another's sessions.
- Added per-user rate limits, size caps, sanitized error responses, and log redaction of session identifiers to limit abuse and data exposure.

AI Sustainable Labs: Governed RAG Assistant [\[Demo\]](#) [\[GitHub\]](#) | *Flask, ChromaDB, BM25, Gemini*

Feb – May 2026

- Scored **205/208** on an independently judged 208-question evaluation of the UMass Boston Sustainable Solutions Lab's public assistant, covering correctness, citations and hallucination, with **zero verified defects**: every remaining miss was hand-checked against the source and traced to a judge disagreement, not a wrong answer.
- Implemented data-governance controls: an allowlist on every server-to-browser field so internal evidence cannot leak, de-identified quality-review logs, admin-only aggregate reporting, and pre-retrieval input screening with rate limits.

ClaimSight: Insurance Claim Triage Platform [\[Demo\]](#) [\[GitHub\]](#) | *FastAPI, Firebase Auth, SAM 2, Docker*

June – Sept 2026

- Built a claim intake and triage tool that separates claimant and adjuster roles and requires a second reviewer before a claim is adjusted, and documented its controls, from upload validation to rate limiting and trusted-host allowlists, in a checklist that also records the gaps they do not cover.

F1 Race Strategy Predictor [\[Demo\]](#) [\[GitHub\]](#) | *XGBoost, scikit-learn, FastAPI, React*

Dec 2025 – Mar 2026

- Found that the original benchmark leaked future data and rebuilt evaluation as walk-forward validation across 8 seasons and ~157 races; reported effect sizes in standard errors and shipped the simpler model where the difference was not distinguishable from noise.

EXPERIENCE

VITA Program (IRS Volunteer Income Tax Assistance)

Boston, MA

IRS-Certified Tax Preparer

Sept 2021 – May 2023

- Earned IRS VITA certification and prepared **30+ federal returns** for low-income filers, reviewing client source documents for completeness and accuracy and clearing every return through the program's required quality review before filing.

Gyu-Kaku Japanese BBQ

Boston, MA

Server

July 2024 – Present

- Serve **40+ guests per shift** averaging **\$1,200 in sales** while working 20+ hours weekly alongside a full course load, communicating clearly under high-volume pressure.

VietAID

Boston, MA

Youth Leader

July – Sept 2022

- Designed and administered mental health surveys for **200+ Boston Public School students**, analyzed responses, and presented findings and recommendations to school officials.